* May switch lights off even when a person is sitting still.

* Cannot independently balance comfort, energy consumption and security.

Conclusion :-

Intelligent agents enable a smart home to react to environmental conditions and make increasingly sophisticated decisions. Simple reflex agents provide fast response, model-based agents maintain context, goal-based agents plan towards desired states, and utility-based agents balance competing objectives.

CO1 - Assignment T-3.

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AI.

1. Intelligent Agents in a Smart Home Automation System.

A Smart home automation System is an intelligent System that automatically Controls lighting, temperature, and Security by analyzing real-time inputs environmental Conditions. An intelligent agent perceives its environment through Sensors and peufoums actions throught actuators to achieve desired outcomes.

Working principle:

Sensors and Users inputs
↓
Intelligent Agent.
↓
Actuators and Actions

Limitations:

* Cannot remember pucurious States or learn user pref

* The robot can update its internal maze map as it explores.

Conclusion:-

UCS is suitable for finding the least-cost path, while IDS is suitable when memory is limited. A model-based, goal-based agent helps the robot remember the maze and make effective decisions, improving its chances of reaching the exit successfully.

J. Ishwarth.

bot Maza Solving Using Uninformed Search

Strategies:

Uniform Cost Search (UCS): Explores the path with the lowest total cost first. It is useful when different paths have different movement costs. It finds the least cost path if the cost are non-negative.

Iterative Deeping Search (IDS):

Searches the maza repeatedly increasing the depth limit each time. It uses less memory and can find the exit without knowing its depth beforehand.

Limitations:

* The model-based agent remembers explored area and obstacles.
* The goal-based agent focuses on reaching the exit
* UCS Compares path Costs and selections the Lowest-Cost route.